

APO-HYDROCHLOROTHIAZIDE

Hydrochlorothiazide tablets USP 25mg and 50mg

Diuretic Antihypertensive

Pharmacology:

Hydrochlorothiazide inhibits reabsorption of sodium and chloride in the distal tubule thus promoting water loss. The higher urine volume increases potassium loss, therefore restricting salt intake decreases potassium loss. Oral doses are well absorbed and reach peak effect in about 4 hours, with a 6 to 12 hours duration. It is excreted unchanged in the urine with a half-life of 3 to 5 hours. The mild blood pressure reducing effects are initially due to volume reduction but the persisting effect includes other undetermined mechanisms that reduce peripheral resistance. A high salt intake reverse its antihypertensive effect.

Non-melanoma skin cancer: Based on available data from epidemiological studies, cumulative dose-dependent association between hydrochlorothiazide and NMSC has been observed. One study included a population comprised of 71,533 cases of BCC and of 8,629 cases of SCC matched to 1,430,833 and 172,462 population controls, respectively. High hydrochlorothiazide use ($\geq 50,000$ mg cumulative) was associated with an adjusted OR of 1.29 (95% CI: 1.23-1.35) for BCC and 3.98 (95% CI: 3.68-4.31) for SCC. A clear cumulative dose response relationship was observed for both BCC and SCC. Another study showed a possible association between lip cancer (SCC) and exposure to hydrochlorothiazide: 633 cases of lip-cancer were matched with 63,067 population controls, using a risk-set sampling strategy. A cumulative dose-response relationship was demonstrated with an adjusted OR 2.1 (95% CI: 1.7-2.6) increasing to OR 3.9 (3.0-4.9) for high use ($\sim 25,000$ mg) and OR 7.7 (5.7-10.5) for the highest cumulative dose ($\sim 100,000$ mg).

Indications:

Adjunctive therapy in edema associated with congestive heart failure, hepatic cirrhosis, corticosteroid and estrogen therapy and in edema of renal origin (i.e. nephrotic syndrome, acute glomerulonephritis, chronic renal disease). In the management of hypertension, hydrochlorothiazide may be used alone or as an adjunct to other antihypertensive drugs. Since it enhances the action of these agents, their dosage must be reduced to avoid an excessive drop in blood pressure and other adverse effects.

Contraindications:

Anuria; discontinue if increasing azotemia and oliguria occur during treatment of severe progressive renal disease. Do not use in patients known to be sensitive to thiazides or other sulfonamide derived drugs.

Warnings & Precautions:

General

Patients on long therapy with hydrochlorothiazide are required to be on potassium rich diet. Periodic determinations of serum electrolytes to detect possible electrolyte imbalance should be performed.

The antihypertensive effects of the drug may be enhanced in the postsympathectomy patient.

Non-melanoma skin cancer

An increased risk of non-melanoma skin cancer (NMSC) [basal cell carcinoma (BCC) and squamous cell carcinoma (SCC)] with increasing cumulative dose of hydrochlorothiazide exposure has been observed in two epidemiological studies based on the Danish National Cancer Registry. Photosensitizing actions of hydrochlorothiazide could act as a possible mechanism for NMSC. Patients taking hydrochlorothiazide should be informed of the risk of NMSC and advised to regularly check their skin for any new lesions and promptly report any suspicious skin lesions. Possible preventive measures such as limited exposure to sunlight and UV rays and, in case of exposure, adequate protection should be advised to the patients in order to minimize the risk of skin cancer. Suspicious skin lesions should be promptly examined potentially including histological examinations of biopsies. The use of hydrochlorothiazide may also need to be reconsidered in patients who have experienced previous NMSC.

Endocrine and Metabolism

- Calcium: Calcium excretion is decreased by thiazides.
- Chloride: Chloride deficiency is generally mild and does not require specific treatment except under special conditions such as renal or/and hepatic disease.
- Dilutional Hyponatremia: Dilutional hyponatremia may occur in edematous patients in hot weather; appropriate

therapy is water restriction rather than administration of salt except when hyponatremia is life threatening. In actual salt depletion, appropriate replacement is the therapy of choice.

- Fluid and electrolyte imbalance: All patients receiving thiazide should be observed for clinical signs of fluid or electrolyte imbalance: namely hyponatremia, hypochloremic alkalosis, and hypokalemia. Serum and urine electrolyte determinations are particularly important when the patient is vomiting excessively, or receiving parenteral fluids. Warning signs of serum electrolyte imbalance, irrespective of cause are: dryness of mouth, thirst, weakness, lethargy, drowsiness, restlessness, muscle pains or cramps, muscular fatigue, hypotension, oliguria, tachycardia, and gastrointestinal disturbances such as nausea and vomiting. Serum electrolytes may also be influenced by medication such as digitalis.
- Hyperuricemia: Hyperuricemia may occur or frank gout may be precipitated in certain patients receiving thiazide therapy.
- Hypokalemia: Hypokalemia may develop, especially with rapid diuresis, when severe cirrhosis is present or during concomitant use of corticosteroids or ACTH. Deficient oral electrolyte intake will also contribute to hypokalemia. Hypokalemia may sensitize or exaggerate the response of the heart to the toxic effects of digitalis (e.g. increased ventricular irritability). Hypokalemia may be avoided or treated by the use of potassium supplements.
- Insulin: Insulin requirements in diabetic patients may be increased, decreased, or remain unchanged. Latent diabetes mellitus may become manifest during thiazide therapy. Concomitant therapy with lithium is not recommended with diuretics because of the reduction of renal clearance of lithium and therefore an added risk of lithium toxicity.
- Parathyroid gland: Pathological changes in the parathyroid gland with hypercalcemia and hypophosphatemia have been observed in a few patients on prolonged thiazide therapy. The common complications of hyperparathyroidism such as renal lithiasis, bone resorption, and peptic ulceration have not been reported. Use of thiazides should be discontinued before carrying out tests for parathyroid function.
- Protein bound iodine (PBI): Thiazides may decrease serum PBI levels without signs of thyroid disturbance.

Gastrointestinal

Non-specific small bowel lesions consisting of stenosis with or without ulceration, may occur in association with the administration of enteric coated potassium salts, alone or with oral diuretics. These small bowel lesions have caused obstruction, hemorrhage and perforation. Surgery was frequently required and deaths have occurred. Available information tends to implicate enteric coated potassium salts, although lesions of this type also occur spontaneously. Such preparations should be used only when adequate dietary supplementation is not practical, and should be discontinued immediately if abdominal pain, distention, nausea, vomiting or gastrointestinal bleeding occur.

Hepatic/Biliary/Pancreatic

APO-HYDROCHLOROTHIAZIDE should be used with caution in patients with impaired hepatic function or progressive liver disease, since minor alterations of fluid and electrolyte balance or of serum ammonia may precipitate hepatic coma.

Immune

The possibility of exacerbation or activation of systemic lupus erythematosus has been reported. Sensitivity reactions may occur in patients with or without a history of allergy or bronchial asthma.

Acute Myopia and Secondary Angle-Closure Glaucoma

Hydrochlorothiazide, a sulfonamide, can cause an idiosyncratic reaction, resulting in acute transient myopia and acute angle-closure glaucoma. Symptoms include acute onset of decreased visual acuity or ocular pain and typically occur within hours to weeks of drug initiation. Untreated acute angle-closure glaucoma can lead to permanent vision loss. The primary treatment is to discontinue hydrochlorothiazide as rapidly as possible. Prompt medical or surgical treatments may need to be considered if the intraocular pressure remains uncontrolled. Risk factors for developing acute angle-closure glaucoma may include a history of sulphonamide or penicillin allergy.

Photosensitivity

Photosensitivity reactions have been reported with the use of thiazide diuretics.

If the photosensitivity reactions occur during treatment with hydrochlorothiazide-containing drugs, termination of the treatment is recommended.

Renal

In progressive renal impairment, therapy with Apo-Hydrochlorothiazide should be withheld or discontinued. Hydrochlorothiazide may commence or precipitate azotemia. It should be used with caution in patients with severely impaired renal function to avoid toxic or cumulative effect. If azotemia becomes more severe and oliguria occurs during treatment of patients with severe renal disease, administration of the diuretic must be stopped.

Acute Respiratory Toxicity

Very rare severe cases of acute respiratory toxicity, including acute respiratory distress syndrome (ARDS) have been reported after taking hydrochlorothiazide. Pulmonary oedema typically develops within minutes to hours after hydrochlorothiazide intake. At the onset, symptoms include dyspnoea, fever, pulmonary deterioration and hypotension. If diagnosis of ARDS is suspected, Apo-Hydrochlorothiazide should be withdrawn and appropriate treatment given. Hydrochlorothiazide should not be administered to patients who previously experienced ARDS following hydrochlorothiazide intake.

Pregnancy:

The routine use of thiazide diuretics in an otherwise healthy pregnant woman with or without edema is contraindicated. Thiazides cross the placental barrier and may carry a similar risk to the fetus of experiencing adverse effects such as jaundice, thrombocytopenia, etc. Edema in pregnancy, resulting from restriction of venous return by the expanded uterus, is treated through elevation of the lower extremities and use of support hose. In a small percentage, more generalized edema may occur and also does not indicate use of diuretics. Rarely, gross edema giving extreme discomfort and due to no other apparent cause may be treated with a short course of diuretics if the above methods and salt restriction fail. Pathological edema such as cardiac, nephrotic or hepatic edema may indicate use of diuretics. In hypertension, peripheral vasodilators should be used first, but diuretics may be added if blood pressure control is not adequate in the presence of excessive edema. This also applies to toxemia. Studies of diuretics for preventing toxemia and improving fetal survival in hypertension, excessive edema or proteinuria have been conflicting, as is the evidence for improved fetal survival. Where a woman has been well established on a thiazide diuretic for hypertension prior to pregnancy, its withdrawal may not be necessary. As thiazides increase blood uric acid concentration, levels should be taken before and during pregnancy but their value in assessing the onset of toxemia may still be lost.

Lactation:

Thiazides are excreted into the milk of nursing women, although apparently not in significant amounts. The potential for idiosyncratic or allergic reactions in the infant should be considered. It should be noted that diuretics may partially inhibit lactation.

Drug Interactions:

Hydrochlorothiazide adds to or potentiates the action of other antihypertensive drugs. Potentiation occurs especially with ganglionic or peripheral adrenergic blocking drugs.

Drug-Drug Interactions:

Proper Name	Ref.	Effect	Clinical comment
Norepinephrine	C	Thiazides may decrease arterial responsiveness to norepinephrine. This diminution is not sufficient to preclude effectiveness of the pressor agent for therapeutic use.	
Alcohol, barbiturates, or narcotics	C	Potentiation of orthostatic hypotension may occur.	Avoid alcohol, barbiturates or narcotics, especially with initiation of therapy.
Amphotericin B	T	Amphotericin B increases the risk of hypokalemia induced by thiazide diuretics.	Monitor serum potassium level.
Antidiabetic agents (e.g. insulin and oral hypoglycemic agents)	CT	Thiazide-induced hyperglycemia may compromise blood sugar control. Depletion of serum potassium augments glucose intolerance.	Monitor glycemic control, supplement potassium if necessary, to maintain appropriate serum potassium levels, and adjust diabetes medications as required.

Proper Name	Ref.	Effect	Clinical comment
Antihypertensive drugs	CT	Hydrochlorothiazide may potentiate the action of other antihypertensive drugs (e.g. guanethidine, methyldopa, beta- blockers, vasodilators, calcium channel blockers, ACEI, ARB, and direct renin inhibitors).	
Antineoplastic drugs, including cyclophosphamide and methotrexate	C	Concomitant use of thiazide diuretics may reduce renal excretion of cytotoxic agents and enhance their myelosuppressive effects.	Hematological status should be closely monitored in patients receiving this combination. Dose adjustment of cytotoxic agents may be required.
Bile acid sequestrants, eg. cholestyramine	CT	Bile acid sequestrants bind thiazide diuretics in the gut and impair gastrointestinal absorption by 43 to 85%. Administration of thiazide 4 hours after a bile acid sequestrant reduced absorption of hydrochlorothiazide by 30 to 35%.	Give thiazide 2 to 4 hours before or 6 hours after the bile acid sequestrant. Maintain a consistent sequence of administration. Monitor blood pressure, and increase dose of thiazide, if necessary.
Calcium and vitamin D supplements	C	Thiazides decrease renal excretion of calcium and increase calcium release from bone.	Monitor serum calcium, especially with concomitant use of high doses of calcium supplements. Dose reduction or withdrawal of calcium and/or vitamin D supplements may be necessary.
Carbamazepine	C	Carbamazepine may cause clinically significant hyponatremia. Concomitant use with thiazide diuretics may potentiate hyponatremia.	Monitor serum sodium levels. Use with caution.
Corticosteroids, and adrenocorticotrophic hormone (ACTH)	T	Intensified electrolyte depletion, particularly hypokalemia, may occur.	Monitor serum potassium, and adjust medications, as required.
Digoxin	CT	Thiazide-induced electrolyte disturbances, i.e. hypokalemia, hypomagnesemia, increase the risk of digoxin toxicity, which may lead to fatal arrhythmic events.	Concomitant administration of hydrochlorothiazide and digoxin requires caution. Monitor electrolytes and digoxin levels closely. Supplement potassium or adjust doses of digoxin or thiazide, as required. Dose adjustment of thiazide may be required.
Drugs that alter GI motility, i.e., anti-cholinergic agents, such as atropine and prokinetic agents, such as metoclopramide, domperidone	CT, T	Bioavailability of thiazide diuretics may be increased by anticholinergic agents due to a decrease in gastrointestinal motility and gastric emptying. Conversely, prokinetic drugs may decrease the bioavailability of thiazide diuretics.	Dosage adjustment of thiazides may be required
Gout medications (allopurinol, uricosurics, xanthine oxidase	T, RC	Thiazide-induced hyperuricemia may compromise control of gout by allopurinol and probenecid. The co-administration of	Dosage adjustment of gout medications may be required.

Proper Name	Ref.	Effect	Clinical comment
inhibitors)		hydrochlorothiazide and allopurinol may increase the incidence of hypersensitivity reactions to allopurinol.	
Lithium	CT	Thiazide diuretics reduce the renal clearance of lithium and add a high risk of lithium toxicity.	Concomitant use of thiazide diuretics with lithium is generally not recommended. If such use is deemed necessary, reduce lithium dose by 50% and monitor lithium levels closely.
Nonsteroidal anti-inflammatory drugs (NSAID)	CT	NSAID-related retention of sodium and water antagonises the diuretic and antihypertensive effects of thiazides. NSAID-induced inhibition of renal prostaglandins leading to decreases of renal blood flow, along with thiazide-induced decreases in GFR may lead to acute renal failure. Patients with heart failure may be at particular risk.	If combination use is necessary, monitor renal function, serum potassium, and blood pressure closely. Dose adjustments may be required.
Selective serotonin reuptake inhibitors (SSRIs, e.g. citalopram, escitalopram, sertraline)	T, C	Concomitant use with thiazide diuretics may potentiate hyponatremia.	Monitor serum sodium levels. Use with caution.
Skeletal muscle relaxants of the curare family, e.g., tubocurarine	C	Thiazide drugs may increase the responsiveness of some skeletal muscle relaxants, such as curare derivatives	
Topiramate	CT	Additive hypokalemia. Possible thiazide-induced increase in topiramate serum concentrations.	Monitor serum potassium and topiramate levels. Use potassium supplements, or adjust topiramate dose as necessary.

Legend: C= Case Study; RCS =Retrospective Cohort Study; CT = Clinical Trial; T =Theoretical

Adverse Effects:

Whenever adverse reactions are moderate or severe, thiazide dosage should be reduced or therapy withdrawn.

Cardiovascular: Orthostatic hypotension (may be aggravated by alcohol, barbiturates, or narcotics).

Central nervous system: Dizziness, vertigo, paresthesias, headache, xanthopsia.

Gastrointestinal system: Anorexia, gastric irritation, nausea, vomiting, cramps, diarrhea, constipation, jaundice (intrahepatic cholestatic jaundice), pancreatitis, sialadenitis.

Hematologic: Leukopenia, agranulocytosis, thrombocytopenia, aplastic anemia.

Hypersensitivity: Purpura, photosensitivity, rash, urticaria, necrotizing angitis (vasculitis), fever, respiratory distress including pneumonitis, anaphylactic reactions.

Other: Hyperglycemia, glycosuria, hyperuricemia, muscle spasm, weakness, restlessness, transient blurred vision.

Eye disorders:

Frequency 'not known': Choroidal effusion, acute myopia, acute angle-closure glaucoma

Neoplasms benign, malignant and unspecified (incl cysts and polyps)

Frequency 'not known': Non-melanoma skin cancer (Basal cell carcinoma and Squamous cell carcinoma)

Based on available data from epidemiological studies, cumulative dose-dependent association between hydrochlorothiazide and NMSC has been observed.

Respiratory, thoracic and mediastinal disorders

Frequency 'very rare': Acute respiratory distress syndrome (ARDS)

Overdose:**Symptoms**

Overdosage may lead to excessive diuresis with electrolyte depletion (hypokalemia, hypochloremia, hyponatremia) and dehydration. Signs are dry mouth, thirst, weakness, lethargy, drowsiness, restlessness muscle pains or cramps, muscular fatigue, hypotension, oliguria tachycardia, gastrointestinal disturbances, mental confusion, delirium, convulsions, shock, coma. If digitalis has also been administered, hypokalemia may accentuate myocardial abnormalities (e.g., cardiac arrhythmias). Hydrochlorothiazide may precipitate hepatic coma in cirrhotics, potentiate other antihypertensive agents, and decrease responsiveness to norepinephrine.

Treatment

There is no specific antidote. If ingestion is recent, gastric lavage or emesis may reduce absorption; when ingestion has been earlier, infusions may be helpful to promote urinary excretion. Otherwise, management includes symptomatic treatment with special attention to cardiac rate and output, blood volume, electrolyte balance, dehydration, paralytic ileus, urinary function, hepatic coma, and cerebral activity. Administration of sympathomimetic drugs (e.g., dopamine) may be indicated. Administer oxygen or artificial respiration for respiratory impairment.

Route of Administration:

Oral

Dosage:

Diuresis: the usual initial adult dose is 25 to 200 mg per day; the maintenance dose is 25 to 100 mg per day, depending on patient response. Some patients may respond to intermittent therapy (alternate days or 3 to 5 days per week). The usual oral dosage for children is 2 mg/kg per day, given in 2 divided doses. Infants under 6 months of age may require up to 3 mg/kg per day, in 2 divided doses.

Hypertension: Usual adult dose is 25 to 50 mg once or twice daily. Doses above this level may increase effectiveness very little but cause more side effects such as hypokalemia. In hypertension associated with renal failure volume overload more potent diuretics, such as loop diuretics may have to be used. On addition of a thiazide to a regimen of other nondiuretic antihypertension agents, an additive blood pressure lowering effect can be anticipated.

Availability of Dosage Form:**Apo-Hydrochlorothiazide 25 mg**

Product description: Pale pink, round, flat-faced, bevelled-edged tablet. Scored & engraved APO over 25 on one side, other side plain. Available in HDPE bottles of 30's, blister pack of 100's or 500's. Not all pack sizes are marketed.

Apo-Hydrochlorothiazide 50mg

Product description: Pale pink, round, flat-faced, bevelled-edged tablet. Scored & engraved APO over 50 on one side, other side plain. Available in HDPE bottles of 30's, blister pack of 100's or 500's. Not all pack sizes are marketed.

Storage Conditions

Store below 30°C.

Manufacturer: Apotex Inc, 50 Steinway Boulevard, Etobicoke, Ontario, Canada, M9W 6Y3.

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